EXERCISES – 1

1. Which of the following systems of linear equations are inconsistent :

(i) $\left.\begin{aligned} 2x_1 + 3x_2 &= 1 \\ 4x_1 + 6x_2 &= 3 \end{aligned}\right\}$

(ii) $\left.\begin{aligned} x_1 + x_2 &= 1 \\ x_1 + 2x_2 &= 3 \\ x_1 - x_2 &= 2 \end{aligned}\right\}$

(iii) $\left.\begin{aligned} x_1 + x_2 + x_3 &= 1 \\ 2x_1 + 2x_2 + 2x_3 &= 1 \\ 3x_1 + 3x_2 + 3x_3 &= 2 \end{aligned}\right\}$

(iv) $\left.\begin{aligned} x_1 + 3x_2 + 2x_3 &= 7 \\ 2x_1 + x_2 + 3x_3 &= 8 \\ 3x_1 + 4x_2 + 6x_3 &= 16 \\ 6x_1 + 8x_2 + 11x_3 &= 20 \end{aligned}\right\}$

Answers:

(i) Inconsistent, (ii) Inconsistent, (iii) Inconsistent.

(iv) Inconsistent

2. Which of the following systems of linear equations are consistent? Find all solutions of the consistent system.

(i) $\left.\begin{aligned} 5x_1 + 3x_2 - 3x_3 &= -1 \\ 3x_1 + 2x_2 - 2x_3 &= -1 \\ 2x_1 - x_2 + 2x_3 &= 8 \end{aligned}\right\}$

(ii) $\left.\begin{aligned} 3x_1 - x_2 + 7x_3 &= 0 \\ 2x_1 - x_2 + 4x_3 &= \frac{1}{2} \\ x_1 - x_2 + x_3 &= 1 \\ 6x_1 - 4x_2 + 10x_3 &= 3 \end{aligned}\right\}$

Answers : (i) $x_1 = 1, x_2 = 2, x_3 = 4$

(ii) $x_1 = -3a - \frac{1}{2}, x_2 = -2a - \frac{3}{2}, x_3 = a.$

where a is any real number.

3. Express the following systems of linear equations in echelon form and solve them :

(i) $\left.\begin{aligned} x_1 - x_2 + 2x_3 + x_4 &= 0 \\ -x_1 + 3x_3 + 2x_4 &= 2 \\ 2x_1 + x_2 - x_4 &= 1 \\ 2x_1 + 2x_2 + x_3 + 3x_4 &= 14 \end{aligned}\right\}$

(ii) $\left.\begin{aligned} x_1 + 2x_2 + x_3 &= -1 \\ 6x_1 + x_2 + x_3 &= -4 \\ 2x_1 - 3x_2 - x_3 &= 0 \\ -x_1 - 7x_2 - 2x_3 &= 7 \\ x_1 - x_2 &= 1 \end{aligned}\right\}$

Answers : (i) $x_1 = 1, x_2 = 2, x_3 = -1, x_4 = 3.$

(ii) $x_1 = -1, x_2 = -2, x_3 = 4.$

(ii) Use echelon method to solve the following system of linear equations :

$$\left.\begin{array}{rrrr} x_1 + x_2 & & + 3 = 0 \\ 2x_1 - 2x_2 - x_3 & & + 8 = 0 \\ 4x_1 & - x_3 & + 14 = 0 \\ x_1 - 3x_2 - x_3 & & + 5 = 0 \end{array}\right\}$$ [D. U. S. 1984]

Answers : (i) $x_1 = \frac{228}{2679}, x_2 = \frac{2698}{2679}, x_3 = \frac{29}{141},$

(ii) $x_1 = \frac{a - 14}{4}, x_2 = \frac{2-a}{4}, x_3 = a$ where a is any real number.

10. (i) Solve the following linear equations :

$$\left.\begin{array}{rrrrr} x_1 + x_2 + 2x_3 & + x_4 & = & 5 \\ 3x_1 + 2x_2 - x_3 & + 3x_4 & = & 6 \\ 4x_1 + 3x_2 + x_3 & + 4x_4 & = & 11 \\ 2x_1 + x_2 - 3x_3 & + 2x_4 & = & 1 \end{array}\right\}$$

(ii) Find all rational solutions of the following system of inear equations :

$$\left.\begin{array}{rrrrr} x_1 + x_2 & - x_3 & - x_4 & = & -1 \\ 3x_1 + 4x_2 - x_3 & - 2x_4 & = & 3 \\ x_1 + 2x_2 + x_3 & & = & 5 \end{array}\right\}$$

Answers : (i) $x_1 = 5a - b - 4, x_2 = -7a + 9, x_3 = a, x_4 = b$ where a and b are arbitrary real numbers.

(ii) $x_1 = 3a + 2b - 7, x_2 = -2a - b + 6, x_3 = a, x_4 = b$ where a and b are arbitrary real numbers.

11. Determine whether each system has a non-zero olution. If exists, find the non-zero solutions.

(i) $\left.\begin{array}{l} x_1 + x_2 + x_3 = 0 \\ x_1 - x_2 + x_3 = 0 \\ x_1 + x_2 - x_3 = 0 \end{array}\right\}$ (ii) $\left.\begin{array}{rrrr} 3x_1 + 2x_2 + & x_3 & = 0 \\ 7x_1 - x_2 & + 21x_3 & = 0 \\ x_1 + 12x_2 & - 37x_3 & = 0 \end{array}\right\}$

Answers : (i) $x_1 = 0, x_2 = 0, x_3 = 0.$

(ii) $x_1 = -\frac{43}{17}a, x_2 = \frac{56}{17}a, x_3 = a$ where a is any real number.

4. Solve each of the following linear systems of equations :

(i) $\left.\begin{aligned} 2x_1 - 3x_2 &= -2 \\ 2x_1 + x_2 &= 1 \\ 3x_1 + 2x_2 &= 1 \end{aligned}\right\}$

(ii) $\left.\begin{aligned} 4x_1 - 8x_2 &= 12 \\ 3x_1 - 6x_2 &= 9 \\ -2x_1 + 4x_2 &= -6 \end{aligned}\right\}$

(iii) $\left.\begin{aligned} 3x_1 + 2x_2 - x_3 &= -15 \\ 5x_1 + 3x_2 + 2x_3 &= 0 \\ 3x_1 + x_2 + 3x_3 &= 11 \\ 11x_1 + 7x_2 &= -30 \end{aligned}\right\}$

(iv) $\left.\begin{aligned} x + 2y - 3z &= 6 \\ 2x - y + 4z &= 2 \\ 4x + 3y - 2z &= 14 \end{aligned}\right\}$

[D. U. S. 1980, 83]

Answers :

(i) Inconsistent

(ii) $x_1 = 3 + 2a$, $x_2 = a$, where a is any real number.

(iii) $x_1 = -4$. $x_2 = 2$, $x_3 = 7$.

(iv) $x = 2 - a$, $y = 2 + 2a$, $z = a$

where a is any real number.

5. Solve the following systems of linear equations :

(i) $\left.\begin{aligned} 2x_1 + x_2 + 3x_3 &= 0 \\ 3x_1 - 2x_2 + x_3 &= 0 \\ x_1 - 3x_2 - 2x_3 &= 0 \end{aligned}\right\}$

(ii) $\left.\begin{aligned} x_1 - x_2 + x_3 + x_4 &= 0 \\ x_1 + 2x_2 - x_3 - x_4 &= 1 \\ 2x_1 - 2x_2 + x_3 - x_4 &= 0 \\ -3x_2 + x_3 - x_4 &= -1 \end{aligned}\right\}$

Answers : (i) $x_1 = a$, $x_2 = a$, $x_3 = -a$

where a is any real number.

(ii) $x_1 = \frac{2}{3}a + \frac{1}{3}$, $x_2 = -\frac{4}{3}a + \frac{1}{3}$, $x_3 = -3a$, $x_4 = a$

where a is any real number.

6. Find the solution sets of the following systems of linear equations :

(i) $\left.\begin{aligned} x_1 + 2x_2 + x_3 + x_4 &= 6 \\ x_1 - x_2 + x_3 - x_4 &= -2 \\ x_1 + 8x_2 + x_3 + 5x_4 &= 22 \\ 2x_1 + 7x_2 + 2x_3 + 4x_4 &= 20 \end{aligned}\right\}$

(ii) $\left.\begin{aligned} x_1 + 2x_2 + 3x_3 + 4x_4 &= 0 \\ 2x_1 + 3x_2 + 4x_3 &= 1 \\ 3x_1 + 4x_2 + x_4 &= 2 \\ 4x_1 + x_3 + 2x_4 &= 3 \end{aligned}\right\}$

Answers :

(i) $x_1 = \frac{1}{3}(2 + b - 3a)$, $x_2 = \frac{1}{3}(8 - 2b)$, $x_3 = a$, $x_4 = b$

where a and b are arbitrary real numbers.

(ii) $x_1 = \frac{9}{11}$, $x_2 = -\frac{1}{11}$, $x_3 = -\frac{1}{11}$, $x_4 = -\frac{1}{11}$.

7. Which of the following systems of linear equations are consistent? Find all solutions of the consistent system :

(i)
$$\left.\begin{aligned} x_1 - 5x_2 + 4x_3 + x_4 &= 7 \\ 2x_1 + x_2 - 3x_3 - x_4 &= 1 \\ 3x_1 - 4x_2 + x_3 - 5x_4 &= 0 \\ x_1 + x_2 + x_3 + x_4 &= 1 \end{aligned}\right\}$$

(ii)
$$\left.\begin{aligned} 3x_1 + 2x_2 - x_3 + 4x_4 &= 6 \\ -2x_1 + x_2 + 5x_3 + x_4 &= 0 \\ x_1 - 4x_2 + 2x_3 + 8x_4 &= 2 \\ 5x_1 + x_2 - 3x_3 - 2x_4 &= 1 \end{aligned}\right\}$$

Answers :

(i) $x_1 = \frac{64}{55}$, $x_2 = \frac{-69}{55}$, $x_3 = -\frac{28}{55}$, $x_4 = \frac{8}{5}$

(ii) $x_1 = \frac{94}{679}$, $x_2 = \frac{685}{679}$, $x_3 = -\frac{30}{97}$, $x_4 = \frac{79}{97}$.

8. (i) Express the following system of linear equations in echelon form and solve it :

$$\left.\begin{aligned} x + y &= -3 \\ 2x - 2y - z &= -8 \\ 4x - z &= -14 \\ x - 3y - z &= -5 \end{aligned}\right\}$$

[D. U. P.1980]

(ii) Reduce the following system of linear equations to an echelon form and find all solutions :

$$\left.\begin{aligned} x_1 - x_2 + 2x_3 &= 5 \\ 2x_1 + x_2 - x_3 &= 2 \\ 2x_1 - x_2 - x_3 &= 4 \\ x_1 + 3x_2 + 2x_3 &= 1 \end{aligned}\right\}$$

[D. U. P. 1981]
[D. U. S. 1981, 1987]

Answers : (i) $x = \frac{1}{4}(a - 14)$, $y = \frac{2 - a}{4}$, $z = a$

where a is any real number.

(ii) $x_1 = 2$, $x_2 = -1$, $x_3 = 1$.

9. (i) Reduce the system of linear equations

$$\left.\begin{aligned} 5x_1 + 2x_2 - 7x_3 &= 1 \\ 7x_1 - x_2 + 2x_3 &= 0 \\ 2x_1 + 5x_2 - x_3 &= 5 \end{aligned}\right\}$$

[D. U. S. 1982]

into echelon form and hence solve it.

12. Find the non-zero solutions of the following systems of linear homogeneous equations :

(i)
$$\left.\begin{aligned} x_1 + 3x_2 + 2x_3 &= 0 \\ 2x_1 - x_2 + 3x_3 &= 0 \\ 3x_1 - 5x_2 + 4x_3 &= 0 \\ x_1 + 17x_2 + 4x_3 &= 0 \end{aligned}\right\}$$

(ii)
$$\left.\begin{aligned} x_1 - 3x_2 - 2x_3 &= 0 \\ 2x_1 + x_2 + 3x_3 &= 0 \\ 3x_1 - 2x_2 + x_3 &= 0 \end{aligned}\right\}$$

[D. U. P. 1986]

(iii)
$$\left.\begin{aligned} 2x_1 + x_2 + x_3 - 5x_4 &= 0 \\ x_1 + x_2 + x_3 - 4x_4 &= 0 \\ x_1 - x_2 - x_3 + 2x_4 &= 0 \end{aligned}\right\}$$

(iv)
$$\left.\begin{aligned} 2x_1 - x_2 - x_3 + x_4 &= 0 \\ x_1 + 2x_2 - 3x_3 + 3x_4 &= 0 \\ -x_1 - x_2 + 2x_3 - 2x_4 &= 0 \end{aligned}\right\}$$

Answers : (i) $x_1 = -\frac{11}{7}a,\ x_2 = -\frac{a}{7},\ x_3 = a$

where a is any real number.

(ii) $x_1 = -a,\ x_2 = -a,\ x_3 = a$

where a is an arbitrary real number.

(iii) $x_1 = b,\ x_2 = -a + 3b,\ x_3 = a,\ x_4 = b$

where a and b are arbitrary real numbers.

(iv) $x_1 = a - b,\ x_2 = a - b,\ x_3 = a,\ x_4 = b$

where a and b are arbitrary real numbers.

In particular, let $a = 2,\ b = 1$ then $x_1 = 1,\ x_2 = 1,\ x_3 = 2,\ x_4 = 1$ is a particular solution of the system.

13. Find the non-trivial solutions of the following systems of linear homogeneous equations :

(i)
$$\left.\begin{aligned} 2x_1 + 3x_2 - x_3 &= 0 \\ x_1 - x_2 + x_3 &= 0 \\ x_1 + 9x_2 - 5x_3 &= 0 \end{aligned}\right\}$$

(ii)
$$\left.\begin{aligned} x_1 - x_2 + x_3 - x_4 &= 0 \\ 2x_1 + x_2 - x_3 + 2x_4 &= 0 \\ 2x_2 + 3x_3 + x_4 &= 0 \end{aligned}\right\}$$

(iii)
$$\left.\begin{aligned} x_1 - 2x_2 + 3x_3 + x_4 &= 0 \\ x_1 + x_2 - 3x_3 - x_4 &= 0 \\ 2x_1 - x_2 + x_3 - x_4 &= 0 \\ 2x_1 + 2x_2 - 5x_3 - 3x_4 &= 0 \end{aligned}\right\}$$

Answers : (i) $x_1 = -\frac{2a}{5},\ x_2 = \frac{3a}{5},\ x_3 = a$

where a is any real number.

(ii) $x_1 = -\frac{1}{3}a,\ x_2 = -a,\ x_3 = \frac{1}{3}a.\ x_4 = a$

where a is any real number.

(iii) $x_1 = \frac{4a}{3},\ x_2 = \frac{8a}{3},\ x_3 = a,\ x_4 = a$

where a is any real number.

14. Solve the following homogeneous systems of linear equations :

(i) $\left.\begin{aligned} x_1 - 2x_2 + 2x_3 &= 0 \\ 2x_1 + x_2 - 2x_3 &= 0 \\ 3x_1 + 4x_2 - 6x_3 &= 0 \end{aligned}\right\}$

(ii) $\left.\begin{aligned} 3x_1 + 4x_2 + x_3 + 2x_4 + 3x_5 &= 0 \\ 5x_1 + 7x_2 + x_3 + 3x_4 + 4x_5 &= 0 \\ 4x_1 + 5x_2 + 2x_3 + x_4 + 5x_5 &= 0 \\ 7x_1 + 10x_2 + x_3 + 6x_4 + 5x_5 &= 0 \end{aligned}\right\}$

(iii) $\left.\begin{aligned} x_1 + 2x_2 + 2x_3 - x_4 + 3x_5 &= 0 \\ x_1 + 2x_2 + 3x_3 + x_4 + x_5 &= 0 \\ 3x_1 + 6x_2 + 8x_3 + x_4 + 5x_5 &= 0 \end{aligned}\right\}$ **[D. U. H. 1986]**

(iv) $\left.\begin{aligned} x_1 + 2x_2 - 2x_3 + 2x_4 + x_5 &= 0 \\ x_1 + 2x_2 - x_3 + 3x_4 - 2x_5 &= 0 \\ 2x_1 + 4x_2 - 7x_3 + x_4 + x_5 &= 0 \end{aligned}\right\}$ **[D. U. H. 1988]**

Answers : (i) $x_1 = \frac{2a}{5}, x_2 = \frac{6a}{5}, x_3 = a$

where a is any real number.

(ii) $x_1 = -3a - 5b, x_2 = 2a + 3b, x_3 = a, x_4 = 0, x_5 = b$

where a and b are arbitrary real numbers.

(iii) $x_1 = -2a + 5b - 7c, x_2 = a, x_3 = -2b + 2c, x_4 = b, x_5 = c$

where a, b and c are arbitrary real numbers.

(iv) $x_1 = -2a - 4b, x_2 = a, x_3 = -b, x_4 = b, x_5 = 0$

where a and b are arbitrary real numbers.

15. (i) Find a general solution and a particular solution of the following system of homogeneous linear equations :

$$\left.\begin{aligned} x_1 - 3x_2 + x_4 &= 0 \\ 2x_1 + x_2 + 3x_3 - x_4 &= 0 \\ 3x_1 - 17x_2 - 6x_3 + 7x_4 &= 0 \end{aligned}\right\}$$

Answers :

General solution : $x_1 = 0, x_2 = \frac{a}{3}, x_3 = \frac{2a}{9}, x_4 = a$

where a is any real number.

Particular solution : $x_1 = 0, x_2 = 3, x_3 = 2, x_4 = 9.$

(ii) Solve the following homogeneous system of linear equations :

$$\left.\begin{aligned} x_1 - 3x_2 + 4x_3 - x_4 &= 0 \\ 3x_1 + x_2 + 2x_3 + 4x_4 &= 0 \\ 2x_1 - 4x_2 + 6x_3 + x_4 &= 0 \\ 2x_1 + 2x_2 + 2x_4 &= 0 \\ 4x_1 - 4x_2 + 8x_3 &= 0 \end{aligned}\right\}$$

Answer : $x_1 = -a,\ x_2 = x_3 = a,\ x_4 = 0$

where a is an arbitrary real number.

(iii) Solve the following system of linear equations :

$$\left.\begin{aligned} x_1 + x_2 + 2x_3 + x_4 &= 5 \\ 2x_1 + 3x_2 + x_3 - x_4 &= 16 \\ x_1 + 2x_2 - 4x_3 + x_4 &= 2 \\ x_1 + x_2 + 3x_3 - 2x_4 &= 12 \end{aligned}\right\}$$

Answer : $x_1 = 2,\ x_2 = 3,\ x_3 = 1,\ x_4 = -2.$

16. (i) Reduce the following system of linear equations to an echelon form and solve it :

$$\left.\begin{aligned} x + 2y - 3z &= 4 \\ x + 3y + z &= 11 \\ 2x + 5y - 4z &= 13 \\ 2x + 6y + 2z &= 22 \end{aligned}\right\}$$ **[D. U. S. 1980]**

(ii) Solve the following system of linear equations :

$$\left.\begin{aligned} x + 2y - 3z &= 6 \\ 2x - y + 4z &= 2 \\ 4x + 3y - 2z &= 14 \end{aligned}\right\}$$ **[D. U. S. 1980] (Improvement)**

using echelon form or otherwise.

(iii) Solve the following linear equations by using echelon form :

$$\left.\begin{aligned} x + 2y + 2z &= 2 \\ 3x - 2y - z &= 5 \\ 2x - 5y + 3z &= -4 \\ x + 4y + 6z &= 0 \end{aligned}\right\}$$ **[D. U. S. 1983]**

Answers :

(i) $x = 1, y = 3, z = 1$

(ii) $x = 2 - a, y = 2a + 2, z = a$ where a is any real number.

(iii) $x = 2, y = 1, z = -1$.

17. Find the general solution and also a particular solution of the following system of linear equations :

$$\left.\begin{aligned} 2x_1 - x_2 - x_3 + 3x_4 &= 1 \\ 4x_1 - 2x_2 - x_3 + x_4 &= 5 \\ 6x_1 - 3x_2 - x_3 - x_4 &= 9 \\ 2x_1 - x_2 + 2x_3 - 12x_4 &= 10 \end{aligned}\right\}$$ **[D. U. S. 1989]**

Answers : General solution :

$$\begin{cases} x_1 = \frac{1}{2}a + b + 2 \\ x_2 = a \\ x_3 = 5b + 3 \\ x_4 = b \end{cases}$$

where a and b are any two real numbers.

Particular solution : $x_1 = 2, x_2 = 2, x_3 = -2, x_4 = -1$.

18. Solve the following system of linear equations :

$$\left.\begin{aligned} x_1 + x_2 + 2x_4 + 3x_5 &= 5 \\ 2x_1 + 4x_2 - x_3 + 5x_4 + 4x_5 &= -1 \\ x_1 + 3x_2 + 5x_4 + 2x_5 &= -3 \\ 3x_1 + 7x_2 - 3x_3 + 9x_4 + 2x_5 &= -14 \\ 2x_1 + 8x_2 - 4x_3 + 2x_4 + 7x_5 &= -10 \end{aligned}\right\}$$

Answer : $x_1 = 2, x_2 = -3, x_3 = 1, x_4 = 0, x_5 = 2$.

19. Solve the following system of linear equations :

$$\left.\begin{aligned} x + 2y + 3z &= -1 \\ 2x - y - 2z &= 5 \\ 3x + 3y + z &= -2 \\ 4x + 3y + 4z &= 3 \\ -4x + 2y + 4z &= -10 \end{aligned}\right\}$$

Answer : $x = 2, y = -3, z = 1$.

20. Solve the following systems of linear equations

(i) $\left.\begin{aligned} x_1 + 2x_2 &= 1 \\ -x_1 + x_2 &= 0 \\ 2x_1 + 4x_2 &= 3 \end{aligned}\right\}$ (ii) $\left.\begin{aligned} 2x + 3y - 2z &= 5 \\ x - 2y + 3z &= 2 \\ 4x - y + 4z &= 1 \end{aligned}\right\}$ [D.U.P. 1991]

(iii) $\left.\begin{aligned} x_1 + x_2 + x_3 &= 3 \\ x_1 - x_2 + 2x_3 &= -3 \\ 3x_1 - x_2 + 5x_3 &= -2 \\ 2x_1 - x_2 - x_3 &= 4 \end{aligned}\right\}$

Answers : (i) Inconsistent (ii) Inconsistent

(iii) Inconsistent.

21. Find all solutions of the following system of linear equations :

$$\left.\begin{aligned} x_1 + x_2 - 2x_3 + x_4 + 3x_5 &= 1 \\ 2x_1 - x_2 + 2x_3 + 2x_4 + 6x_5 &= 2 \\ 3x_1 + 2x_2 - 4x_3 - 3x_4 - 9x_5 &= 3 \end{aligned}\right\}$$ **[D. U. H. 1989]**

Answer : $x_1 = 1,\ x_2 = 2a,\ x_3 = a,\ x_4 = -3b,\ x_5 = b$

where a and b are arbitrary real numbers.

22. Solve the following linear equations :

$$\left.\begin{aligned} x_1 + 2x_2 - 3x_3 + 4x_4 &= 1 \\ 2x_1 + 5x_2 - 5x_3 + 6x_4 &= 3 \\ x_1 + 4x_2 - x_3 &= 3 \\ 2x_1 + 3x_2 - 7x_3 + 10x_4 &= 1 \end{aligned}\right\}$$ **[D. U. H. 1987]**

Answer : $x_1 = -1 + 5a - 8b,\ x_2 = 1 - a + 2b,\ x_3 = a,\ x_4 = b$

where a and b are arbitrary real numbers.

23. Find all solutions of the following homogeneous systems of linear equations :

(i) $\left.\begin{aligned} x + 2y - 3z &= 0 \\ 2x + 5y + 2z &= 0 \\ 3x - y - 4z &= 0 \end{aligned}\right\}$ **[D. U. H. 1986]**

(ii) $\left.\begin{aligned} x_1 + 4x_2 + 5x_3 + 3x_4 &= 0 \\ 2x_1 + 3x_2 + 5x_3 + x_4 &= 0 \\ 3x_1 + 2x_2 + 5x_3 - x_4 &= 0 \\ 4x_1 + x_2 + 5x_3 - 3x_4 &= 0 \end{aligned}\right\}$ **[D. U. P. 1985]**

(iii)
$$\left.\begin{aligned} x_1 - x_2 - x_3 - x_4 &= 0 \\ x_1 + 3x_2 - x_3 + x_4 &= 0 \\ 3x_1 - 7x_2 - x_3 - 6x_4 &= 0 \\ 2x_1 + 2x_2 - 2x_3 &= 0 \\ 6x_1 - 2x_2 - 4x_3 - 5x_4 &= 0 \end{aligned}\right\}$$ [D. U. P. 1988]

(iv)
$$\left.\begin{aligned} 2x_1 + 2x_2 - x_3 + x_5 &= 0 \\ -x_1 - x_2 + 2x_3 - 3x_4 + x_5 &= 0 \\ x_1 + x_2 - 2x_3 - x_5 &= 0 \\ x_3 + x_4 + x_5 &= 0 \end{aligned}\right\}$$

Answers : (i) $x = y = z = 0$, No non-zero solution.

(ii) $x_1 = b - a$, $x_2 = -(a + b)$, $x_3 = a$ and $x_4 = b$

where a and b are arbitrary real numbers.

(iii) $x_1 = 2a$, $x_2 = -a$, $x_3 = a$, $x_4 = 2a$

where a is arbitrary real number.

(iv) $x_1 = -a - b$, $x_2 = a$, $x_3 = -b$, $x_4 = 0$, $x_5 = b$

where a and b are arbitrary real numbers.

24. Reduce the following system of linear equations to echelon form and solve it :

$$\left.\begin{aligned} x_1 + x_2 + x_3 + 2x_4 + 3x_5 &= 13 \\ -2x_1 - 2x_2 + x_3 - 3x_4 - 4x_5 &= 5 \\ 3x_1 + 2x_2 + x_4 + 5x_5 &= 10 \\ x_1 + x_2 + 2x_3 - x_4 + 9x_5 &= 18 \end{aligned}\right\}$$

Answer : $x_1 = 5 - a$, $x_2 = a$, $x_3 = 11$, $x_4 = 0$, $x_5 = -1$.

where a is any real number.

25. Solve the following system of linear equations :

$$\left.\begin{aligned} x_1 + 2x_2 + 3x_3 + 4x_4 &= 5 \\ 2x_1 + x_2 + 4x_3 + x_4 &= 2 \\ 3x_1 + 4x_2 + x_3 + 5x_4 &= 6 \\ 2x_1 + 3x_2 + 5x_3 + 2x_4 &= 3 \end{aligned}\right\}$$

Answer : $x_1 = \frac{15}{71}$, $x_2 = \frac{-11}{71}$, $x_3 = \frac{10}{71}$, $x_4 = \frac{83}{71}$.

30. Determine the values of λ such that the following system in unknowns x, y and z has (i) a unique solution, (ii) no solution (iii) more than one solution :

$$\left.\begin{aligned} x \quad\quad - 3z &= -3 \\ 2x + \lambda y - \ z &= -2 \\ x + 2y + \lambda z &= 1 \end{aligned}\right\}$$

Answers : (i) $\lambda \neq 2$ and $\lambda \neq -5$, (ii) $\lambda = -5$, (iii) $\lambda = 2$.

31. Determine the values of λ such that the following system of linear equations in unknowns x, y and z has (i) a unique solution (ii) no solution (iii) more than one solution :

$$\left.\begin{aligned} x + \ y + \lambda z &= 1 \\ x + \lambda y + \ z &= \lambda \\ \lambda x + y + \ z &= \lambda^2 \end{aligned}\right\}$$

Answers : (i) $\lambda \neq -2$, $\lambda \neq 1$ (ii) $\lambda = -2$ (iii) $\lambda = 1$.

32. Find out the conditions on α, β and γ so that the following systems of non-homogeneous linear equations has a solution :

(i)
$$\left.\begin{aligned} x + 2y - \ 3z &= \alpha \\ 3x - \ y + \ 2z &= \beta \\ 2x - 10y + 16z &= 2\gamma \end{aligned}\right\}$$

(ii)
$$\left.\begin{aligned} x + 2y - \ 3z &= \alpha \\ 2x + 6y - 11z &= \beta \\ 2x - 4y + 14z &= 2\gamma \end{aligned}\right\}$$

Answer : (i) $2\alpha = \beta - \gamma$ (ii) $5\alpha = 2\beta + \gamma$.

33. Find out the conditions on a, b and c so that the following system of non-homogeneous linear equations is consistent and also solve the system for a = 1, b = 1 and c = −2 :

$$\left.\begin{aligned} -2x + y + \ z &= a \\ x - 2y + \ z &= b \\ x + \ y - 2z &= c \end{aligned}\right\}$$

Answer : Condition for consistent is $a + b + c = 0$ and the general solution is $x = \alpha - 1$, $y = \alpha - 1$, $z = \alpha$ where α is arbitrary real number.

A particular solution is $x = 1$, $y = 1$, $z = 2$.

26. Find a general solution and also a particular solution of the following system of linear equations :

$$\left.\begin{aligned} 2x_1 + 3x_2 - x_3 + 5x_4 - 2x_5 &= 4 \\ x_1 + 2x_2 + x_3 + 3x_4 - x_5 &= -1 \\ 2x_1 + x_2 - 6x_3 + 7x_4 + 3x_5 &= 1 \\ 5x_1 + 11x_2 + 7x_3 + 12x_4 - 10x_5 &= 4 \end{aligned}\right\},$$

Answers : $x_1 = -64 - 21a - 24b$, $x_2 = 39 + 11a + 15b$,

$x_3 = -15 - 4a - 5b$, $x_4 = a$, $x_5 = b$.

where a and b are arbitrary real numbers.

In particular $x_1 = -85$, $x_2 = 50$, $x_3 = -19$, $x_4 = 1$, $x_5 = 0$

27. Reduce the following system of linear equations into echelon form and solve it :

$$\left.\begin{aligned} x_1 + 2x_2 + 3x_3 + 4x_4 + 5x_5 &= 1 \\ 2x_1 + 3x_2 + 4x_3 + 5x_4 + 6x_5 &= -1 \\ 3x_1 + 5x_2 + 6x_3 + 7x_4 + 4x_5 &= 2 \\ 4x_1 + 7x_2 + 10x_3 + 13x_4 + 16x_5 &= 1 \\ 5x_1 + 8x_2 + 9x_3 + 10x_4 + 3x_5 &= 3 \end{aligned}\right\}$$

Answer : $x_1 = -7 - 4b$, $x_2 = 7 + a + 10b$, $x_3 = -2 - 2a - 7b$, $x_4 = a$, $x_5 = b$ where a and b are arbitrary real numbers.

28. Reduce the following system of linear equations into echelon form and solve it :

$$\left.\begin{aligned} 4x_1 + 2x_2 + 5x_3 + 7x_4 + x_5 &= 2 \\ x_1 + x_2 + x_3 + x_4 + 5x_5 &= 1 \\ 2x_1 + 3x_2 + 4x_3 + 5x_4 + 6x_5 &= 1 \\ 3x_1 + 9x_2 + 7x_3 + x_4 + 8x_5 &= 9 \\ 5x_1 + x_2 + x_3 + 6x_4 + x_5 &= 0 \end{aligned}\right\}$$

Answer : $x_1 = 1$, $x_2 = 0$, $x_3 = 1$, $x_4 = -1$, $x_5 = 0$.

29. Determine the values of λ such that the following system in unknowns x, y and z has

(i) a unique solution, (ii) no solution, (iii) more than one solution :

$$\left.\begin{aligned} \lambda x + y + z &= 1 \\ x + \lambda y + z &= 1 \\ x + y + \lambda z &= 1 \end{aligned}\right\}$$ **[R. U. H 1985]**

Answers : (i) $\lambda \neq 1$ and $\lambda \neq -2$. (ii) $\lambda = -2$. (iii) $\lambda = 1$.